

Determine the (a) first degree polynomial, (b) the second degree polynomial, that best approximates x^4 in $L_2(w, I)$, where $w(x) = x$ and $I = [0, 1]$.

Solution:

For this setup, the scalar product is defined as

$$(u | v) = \int_0^1 u(x)v(x)w(x) \, dx = \int_0^1 u(x)v(x)x \, dx$$

Find an orthogonal base

$$\varphi_0 = 1$$

$$(\varphi_0 | \varphi_0) = \int_0^1 x \, dx = \frac{1}{2}$$

$$\psi_0 = \frac{\varphi_0}{(\varphi_0 | \varphi_0)} = 2$$

$$\varphi_1 = x - P_{[\varphi_0]}x = x - \frac{(\varphi_0 | x)}{(\varphi_0 | \varphi_0)}\varphi_0 = x - 2 \int_0^1 x^2 \, dx = x - \frac{2}{3}$$

$$(\varphi_1 | \varphi_1) = \int_0^1 \left(x - \frac{2}{3}\right)^2 x \, dx = \frac{1}{36}$$

$$\psi_1 = \frac{\varphi_1}{(\varphi_1 | \varphi_1)} = 36x - 24$$

Now $v \in \mathcal{M} = \{2, 36x - 24\}$

Are looking for

$$\int_0^1 (x^4 - a\psi_0 - b\psi_1)^2 x \, dx$$

From which we can calculate

$$a\psi_0 = \frac{(\psi_0 | x^4)}{(\psi_0 | \psi_0)}\psi_0$$

$$a = \frac{(\psi_0 | u)}{(\psi_0 | \psi_0)}$$

$$(\psi_0 | u) = \int_0^1 2x^5 \, dx = \frac{1}{3}$$

$$(\psi_0 | \psi_0) = \int_0^1 4x \, dx = 2$$

$$a = \frac{1}{6}$$

and

$$b\psi_1 = \frac{(\psi_1 | x^4)}{(\psi_1 | \psi_1)}\psi_1$$

$$b = \frac{(\psi_1 | x^4)}{(\psi_1 | \psi_1)}$$

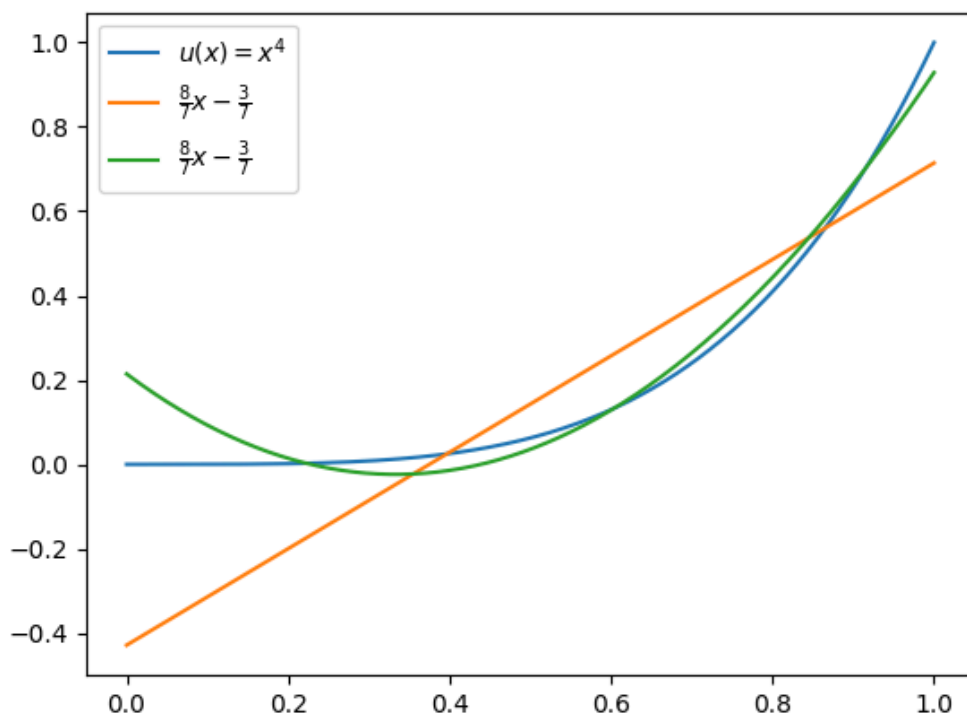
$$(\psi_1 | x^4) = \int_0^1 (36x^6 - 24x^5) dx = \frac{36}{7} - \frac{24}{6} = \frac{8}{7}$$

$$(\psi_1 | \psi_1) = \int_0^1 (1296x^3 - 1728x^2 + 576x) dx = 36$$

$$b = \frac{2}{63}$$

The first degree polynomial that best approximates the function $u(x) = x^4$ in $I = [0, 1]$ is

$$\frac{1}{6}\psi_0 + \frac{2}{63}\psi_1 = \frac{8}{7}x - \frac{3}{7}$$



(b) We are now looking for a second degree polynomial which means our $\psi \in \mathcal{M} = \{1, x, x^2\}$
 ψ_0 and ψ_1 were already calculated in (a) so

$$\begin{aligned}\varphi_2 &= u_2 - P_{[\varphi_0, \varphi_1]} u_2 = x^2 - \frac{(\varphi_0 | x^2)}{(\varphi_0 | \varphi_0)} \varphi_0 - \frac{(\varphi_1 | x^2)}{(\varphi_1 | \varphi_1)} \varphi_1 \\ \frac{(\varphi_0 | x^2)}{(\varphi_0 | \varphi_0)} \varphi_0 &= 2(\varphi_0 | x^2) = 2 \int_0^1 x^3 \, dx = \frac{1}{2} \\ \frac{(\varphi_1 | x^2)}{(\varphi_1 | \varphi_1)} \varphi_1 &= (\varphi_1 | x^2)(36x - 24)\end{aligned}$$

Now calculating

$$(\varphi_1 | x^2) = \int_0^1 \left(x^4 - \frac{2x^3}{3} \right) dx = \frac{1}{30}$$

which gives finally

$$\begin{aligned}\varphi_2 &= x^2 - \frac{6}{5}x + \frac{3}{10} \\ (\varphi_2 | \varphi_2) &= \int_0^1 \left(x^2 - \frac{6x}{5} + \frac{3}{10} \right)^2 x \, dx = \frac{1}{600}\end{aligned}$$

$$\psi_2 = 600x^2 - 720x + 180$$

We can now find the best approximation with

$$\begin{aligned}\int_0^1 (x^4 - a\psi_0 - b\psi_1 - c\psi_2)^2 x \, dx \\ c\psi_2 &= \frac{(\psi_2 | x^4)}{(\psi_2 | \psi_2)} \psi_2 \\ c &= \frac{(\psi_2 | x^4)}{(\psi_2 | \psi_2)} \\ (\psi_2 | x^4) &= \int_0^1 (600x^2 - 720x + 180)x^4 \, dx = \frac{15}{7} \\ (\psi_2 | \psi_2) &= \int_0^1 (600x^2 - 720x + 180)^2 x \, dx = 600 \\ &\downarrow \\ c &= \frac{15}{4200} = \frac{1}{280}\end{aligned}$$

The best second degree polynomial to approximate $u(x) = x^4$ is then

$$a\psi_0 + b\psi_1 + c\psi_2 = \frac{15}{7}x^2 - \frac{10}{7}x + \frac{3}{14}$$

□